

Integration of AI for Detecting Fake Jobs & Networks in Online Recruitment Platforms

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1. Abstract

With more people turning towards freelance platforms like Upwork to find work, the risk of encountering fake job listings and ghost employer networks has become a serious issue. This project introduces a fraud detection system designed to protect job seekers and maintain trust on such platforms. By combining textual analysis and structured job data, the system detects coordinated scam patterns without relying on labeled training data. Through semantic clustering and anomaly detection, it identifies both near-duplicate scams and suspicious employer behavior. The solution helps the platforms flag fraud early, reduce manual moderation, and protect users which ultimately helps in boosting credibility and enabling safer, more transparent online hiring environments for online recruitment platforms.

1. Introduction

In today's day and age, where digitization is becoming a part of human lifecycle, online recruitment platforms e.g. LinkedIn, Indeed and freelancing marketplaces e.g. Upwork, Fiverr, have become one of the best go-to options when it comes to connecting employers with job seekers. While these platforms have opened the doors for endless opportunities and flexible hiring, they are also susceptible to fraudulent activities. These activities can range from posting fake jobs which aims to steal personal information to coordinated networks of deceptive employers, which is becoming sophisticated and harder to detect day by day. In this project, we are using Upwork's job postings data to utilize all the parameters of the jobs posted to determine if they are fake or related to any coordinated scam network, based on the recurring pattern of fake jobs based on their context.

This project expands on the recent study, Fraud-BERT[1] which utilizes labeled textual data of job postings to determine if the jobs posted are fake or not by utilizing structured metadata of the jobs along with textual parameters, to not only just find our fake jobs but also the networks of deceptive job patterns it might belong to.

The proposed model not only uses real time data, but it also involves unsupervised learning using non-labeled dataset, utilizing MiniLM and TabNet Classifier models to understand the textual as well as jobs metadata context,

enabling us to find anomalies, which helps us learn and recognize the fake jobs patterns and detect their network cluster of fake jobs. Job seekers are often unaware when they apply to these illegitimate listings, which can lead to wasted time, emotional stress, and financial loss. By addressing this problem proactively and in real time we are able to maintain trust, safety, and transparency in the online job ecosystem.

2.1. Background and Scope

Online Recruitment Frauds (ORFs) has expanded a lot in the recent years and is not just limited to isolated fake jobs. A lot of these scams now operated through a pattern of deception, which includes posting similar jobs with different names repeatedly, using incomplete company information, or impersonating as some legitimate business/organization. Although awareness has been raised against these job scams, most existing solutions only focuses on detecting individual fake job posts using methods which often fails to keep up with the evolving nature of these scams[4][2] like using predefined rules, basic keyword analysis etc.[5]

This project aims to develop a novel fraud detection framework that does not just stick to single-post classification but go beyond it by identifying patterns across similar job listings to uncover ghost employer networks, which helps in understanding as to how fraudsters behave across multiple listings and various platforms. By studying and utilizing text patterns, job characteristics, and employer behaviors, recruitment fraud can be detected and avoided more effectively ensuring users safety and improving the integrity of digital job platforms.

2.2 Goals

The primary goal of this projects revolves around developing a fraud detection system identify larger scam pattern which is not limited to determining only isolated fake jobs but understanding how ghost employer networks operate on freelancing platforms. It will also be free from using labeled examples or fixed rules but rather explore patterns in job descriptions, employer behavior, and

contextual attributes to identify suspicious activity more holistically.[5]

There are several important implications from a business point of view:

- Protecting users from fraudulent job postings, by identifying them and flagging them more accurately, and preserve their trust in the platform.
- For online recruitment platform, minimizing fraudulent activity increase the chances of attracting more legitimate users, ensuring a safer and more professional environment.
- Enabling proactive risk management will help us detect coordinated fraud early which will allow the platforms to intervene and take measures before any significant harm happens.
- The system is designed such that it can learn from the real-world data using various online recruitment platform which makes it more adaptable to the ever-evolving scam strategies.

Ultimately, the aim of the project is to contribute towards building a more resilient, trustworthy, and intelligent job ecosystem which not only protects users but also ensures that digital hiring remains a force for good rather than exploitation.

2. Background and Literature Survey

One of the studies, that serves as the key motivation for this project is **Fraud-BERT**, a transformer-based deep learning model fine-tuned for fake job detection using contextual information of the job description and integrating it with text classification methods to be able to distinguish between legitimate and fraudulent online recruitment listings. Although, it achieved higher accuracy, it is limited to detecting fraud at a single job post level and not finding interrelations between fake jobs or coordinated scam networks or ghost employers which operates via multiple similar listings. Moreover, it relies heavily on supervised learning with a highly imbalanced dataset and does not utilize other parameters of job postings like, salary, age, position, etc. which can be key for unlocking potential patterns of these scam networks.[1]

This project utilizes contextual features (e.g., employer legitimacy, job title consistency, company size, and online presence) and applying supervised learning models like SVM and Random Forest on textual, structural and Contextual features. Some limitations remain like, the features are manually engineered which restricts adaptability to new fraud patterns. It is limited to Australian dataset lacking global generalization of features and does not explore how language-level patterns across job posts.[2]

This paper focuses on addressing class imbalance in fake job detection by evaluating oversampling and hybrid resampling techniques (SMOTE, ADASYN, Tomek Links) with machine learning models (Decision Trees and Random Forests) on the EMSCAD dataset. Like much other research, it is static and supervised depending on labeled dataset and utilizing SMOTE/ ADASYN to oversample minority class which may not capture the complex patterns of real-world fraudulent behavior.[3]

This paper shows ORF Detector, a machine learning model which utilizes structured features from job postings and uses ensemble classifiers like Random Forest, XGBoost, etc. to detect online recruitment fraud on the EMSCAD dataset. But it lacks textual Semantics i.e. it does not leverage textual embeddings or context-aware language models like BERT.[4]

This paper provides a review of ORF, analyzing its evolution, detection strategies, and socio-technical challenges faced, while aiming for collaborative, policy-driven, and technological solutions to enhance trust. The existing cross-platform collaboration and detection methods are fragmented lacking network-level insights which is key in figuring out ghost job networks. Moreover, Detection of ghost jobs is mostly reactive and struggle to adapt in real-time to evolving fraud tactics.[5]

This paper shows a deep learning approach using BiLSTM to find out sequential patterns in job descriptions and classify whether the job postings is real or fake. The model relies on sequential inputs without context-aware embeddings like BERT and ignores relational features such as employer connections, which limits its ability to capture fraud patterns.[6]

This paper implements deep learning models to identify fraud certifications by analyzing patterns in the data using TabNet. It's sequential attention mechanism helps it to focus on the relevant features, making it effective to detect anomalies without the need for heavy feature engineering.[7]

Our project builds upon the previous Fraud-BERT approach but introduces several key enhancements. We are using Upwork's unlabeled dataset of job postings, uses MiniLM embeddings with unsupervised clustering and reconstruction error analysis to detect fake jobs by identifying semantic and structural patterns amongst various job listings. Contextual and text data combined, have been integrated, leveraging TabNet for structured features, addressing class imbalance using class weights, and finding ghost employer networks beyond individual post-level fraud detection.

Through these improvements, our model not only detects ghost jobs with high accuracy (achieving overall accuracy of 0.84 for TabNet Classifier and silhouette score of 0.63 for MiniLM), but also, for finding out fraud-networks with similar job posting patterns making it a robust

solution for identifying and adapting in real-time to evolving fraud tactics.

3. Ethical Concerns

Even though the goal of this project is to enhance and promote trust and safety in online job platforms, it is also important to make sure that the system is developed and deployed in an ethically responsible and GDPR-compliant manner, with strong safeguards around data privacy, fairness, and transparency.

The project involves using real life job postings and associated employers' information. Even though the data is publicly available, based on the ethical obligation, no personally identifiable information (PII) is exposed or misused, and anonymization & responsible data handling are addressed. When identifying fake jobs one of the major ethical risks involve misclassifying legitimate job postings as fraud, which could unfairly damage the reputation of honest employers, leading to loss of business and trust. Therefore, a mechanism for human oversight before final judgments of the system has been addressed.

If the system is trained and validated on the data that contains geographical, linguistic, or industry-specific biases, it may unfairly flag certain regions, companies, or job categories as suspicious. To ensure fair classification across various demographic, users etc. un-labeled datasets are used to avoid unintended discrimination.

Excessive monitoring or profiling of employer behavior can raise ethical concerns about surveillance and removes trust among legitimate users. This is especially crucial for freelancers and job seekers in developing regions, who are already facing challenges trying to gain visibility or credibility. An extreme fraud detection system may unnecessarily penalize these vulnerable users, reinforcing barriers rather than removing them. Therefore, it's important to design systems that balance protection with fairness and transparency, ensuring that fraud prevention does not come at the cost of trust or opportunity.

By incorporating these safeguards, transparency, and fairness ethical concerns into the system, this project not only aims to detect fraud, but to do it responsibly, protecting all stakeholders while enhancing the integrity of digital job ecosystems.

4. Implementation of Techniques (Models)

To detect fake job postings and uncover the fake employer networks, this project has implemented a hybrid model approach which combines natural language processing, tabular modeling, and unsupervised learning. It involves integration of both textual content analysis and contextual job metadata. This ensure that the model captures both the language-level anomalies and the behavioral and relational

patterns which might represent coordinated fraud. It is implemented in three key stages which are as follows:

5.1 Data Collection and Preprocessing

The dataset used for this project is a real-world collection of Upwork Job Listings which is stored in Google Drive. The pipeline starts with loading the dataset using `pandas.read_csv()` after mounting Google Drive.

Initial cleaning of the data includes:

- Filling missing values in key columns like `Payment_type`, `Client_Country`, `EX_level_demand`.
- Normalizing and cleaning text data from Job Title and Description.
- Concatenating job title and description into a new column `text_input` for embedding.
- Removing records with missing or empty job descriptions.

Additionally, textual content is changed into contextual embeddings using SentenceTransformer with the DistilBERT (MiniLM-L6-v2) model for efficient semantic representation. These embeddings are later used for clustering and similarity detection. Also, individual preprocessing steps for semantic processing and structured contextual processing is shown in the table below.(Table 1)

Phase	Step	Description
Textual Preprocessing for MiniLM-L6-v2	Missing Value Handling	Filled null Job Title and Description with empty strings.
	Text Cleaning & Normalization	Removed newline characters and extra whitespace. Trimmed both title and description fields.
	Textual Consolidation	Created <code>text_input</code> by combining Job Title and Description for a unified semantic input.
	Filtering Incomplete Records	Dropped entries with empty or non-informative <code>text_input</code> .
	Contextual Embedding	Used SentenceTransformer (MiniLM-L6-v2) to generate DistilBERT embeddings of job descriptions.
Structured Preprocessing for TabNet Classifier	Missing Value Imputation	Imputed "Unknown" for missing values in <code>Payment_type</code> , <code>Client_Country</code> , <code>EX_level_demand</code> .
	Categorical Normalization	Preserved categorical values in raw form for later encoding or embedding.
	Feature Consolidation	Retained structured features like budget, duration, client history, and job category.

	Alignment with Text Embeddings	Aligned structured rows with corresponding DistilBERT embeddings for hybrid or comparative modeling.
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5.2 EDA (Exploratory Data Analysis)

Some of the key exploratory analysis which helped in gaining insights into the dataset of Upwork Job Postings are as follows:

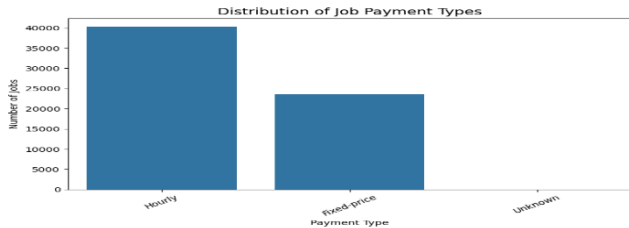


Fig. 1 Payment Type Distribution

The above graph **Fig. 1** shows the distribution of job listings by payment type for Upwork platform, with Hourly jobs being the most common, followed by Fixed price, and very few jobs marked as Unknown.

This helps us identify baseline patterns, so listings with unusual or missing payment types can be flagged as suspicious. In this case, such anomalies may indicate ghost job behavior or low-effort scam posts. It also makes Payment Type a valuable structured feature for fraud detection models.

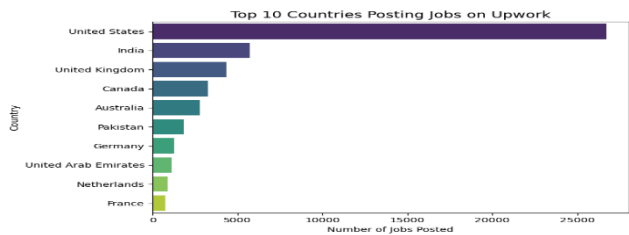


Fig. 2 Geographical Distribution of Jobs Posted

The above graph **Fig. 2** shows the top 10 countries postings of Jobs on Upwork, where Unites States has the highest number of jobs posted, followed by India and then UK. This geographic distribution helps in understanding job origins where suspicious job clusters from unexpected or low-activity regions may indicate ghost employer networks. Country data also provides important contextual signals for the TabNet Classification and unsupervised fraud detection models.

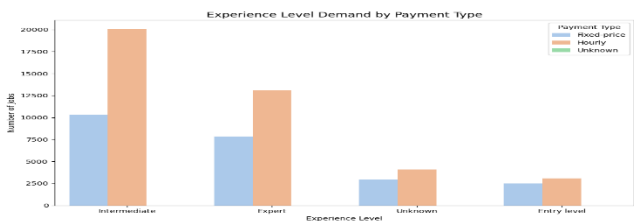


Fig. 3 Experience Demand vs. Payment Type

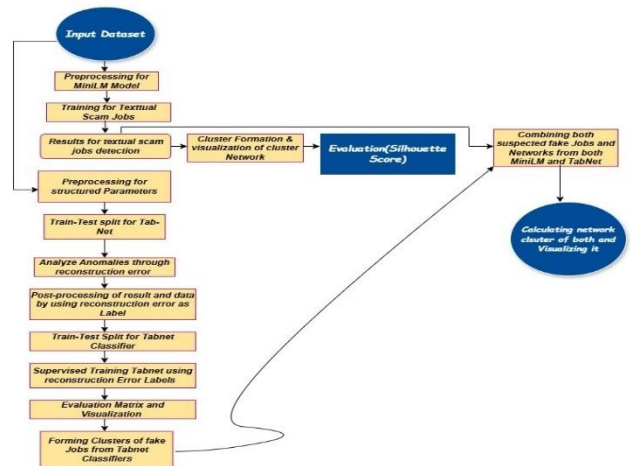
The above graph **Fig. 3** illustrates the demand for different experience levels across payment types. Most jobs target intermediate and expert freelancers, especially for hourly roles. The relatively lower and more ambiguous demand at the entry level or with "unknown" experience types can act as red flags in fraud detection, as scammers often use such less detailed postings to attract unsuspecting applicants. These patterns help enhance fraud detection in our structured analysis.

5.3 Model selection and Training

The project combines both unsupervised and embedding-based approaches rather than just relying on traditional supervised classification.

- **Textual Representation:** Job descriptions are embedded using DistilBERT (MiniLM-L6-v2) to capture semantic context using textual inputs.
- **Similarity based Clustering for Network Detection:** The system uses cosine similarity on DistilBERT embeddings of job descriptions to identify semantically similar job listings. It helps to flags clusters of near-duplicate posts which reveals coordinated scam patterns and ghost job networks.
- **Fraud Scoring:** Using reconstruction error or similarity distance to flag anomalies.
- **Tabular Feature Modeling:** Integrated TabNet (from separate notes) for structured features like payment type, country, budget, etc.
- **Imbalance Handling:** Class weights are used, ensuring GDPR-compliant and interpretable fraud detection and avoiding redundant fake job posts.

5.4 Project Flow



The project flows show the complete working, and the steps involved in detecting fake jobs:

- **Textual Detection:** Preprocess job text and use MiniLM embeddings with cosine similarity to cluster and detect scam jobs which is evaluated via Silhouette Score.
- **Structured Detection:** Clean structured fields and use an autoencoder to detect anomalies, then train a supervised TabNet classifier using reconstruction error labels.
- **Model Evaluation:** Evaluate MiniLM with clustering metrics and TabNet with classification metrics (precision, recall, F1-score) to check fraud detection performance.
- **Fusion & Network Insight:** Merge flagged jobs from both models to form a fraud network, showing coordinated scams across text patterns and structured anomalies.

By combining context-aware embeddings, attention-based modeling, and graph-informed clustering, the system explores methods beyond traditional classification to detect hidden scam ecosystems, making it a novel contribution to recruitment fraud detection.

6.1. The Quantitative Findings

To assess the effectiveness of the proposed system in detecting ghost or fake job postings, below evaluations provide the quantitative performance of the project.

By embedding job descriptions with DistilBERT MiniLM-L6-v2 and applying cosine similarity-based clustering, the model grouped jobs that were textually and contextually similar, which help to determine coordinate ghost job rings, showing similar text reused across multiple listings. The Silhouette Score : 0.5663, suggests clusters are separated in a meaningful way showing distinct semantic patterns.

Evaluation at threshold = 0.5				
	precision	recall	f1-score	support
0	0.96	0.85	0.90	5963
1	0.48	0.80	0.60	1037
accuracy			0.84	7000
macro avg	0.72	0.83	0.75	7000
weighted avg	0.89	0.84	0.86	7000

Fig. 4 Classification Report

Using structured features such as client country, payment type, experience level, and post frequency, the TabNet model was trained to classify for fake jobs. In **Fig. 4** we can see the high Recall (Fraud Class): 0.80, shows that the model can catch most of the fake jobs, whereas the slightly lower precisions Precision (Fraud Class): 0.48, shows some false positives which is expected to happen as we are utilizing semi-supervised learning for TabNet Classifier, which produces imbalanced label for fake jobs which is

handled by class weights. For platforms which prioritize safety and trust, this is acceptable since it minimizes the risk of scam exposure to job seekers, especially new and vulnerable freelancers.

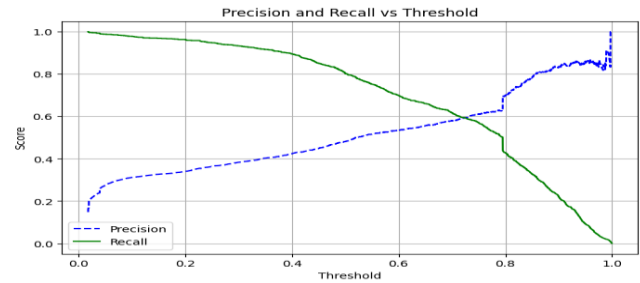


Fig. 5 Precision & Recall vs Threshold

The precision-recall vs threshold curve graph above **Fig. 5** shows the trade-off between detecting more fraud (recall) and reducing false positives (precision). A balanced threshold (0.5–0.6) helps the platforms to fine-tune detection sensitivity based on the risk level and region.

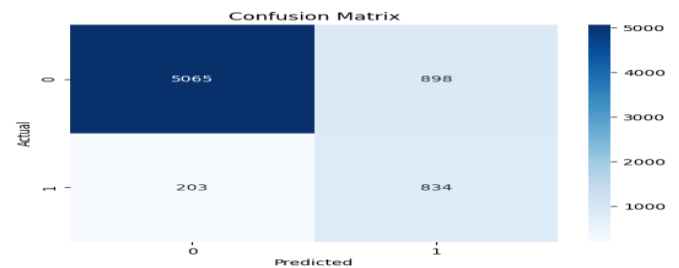


Fig. 6 Confusion Matrix

The confusion matrix in the above diagram **Fig. 6** shows strong accuracy for detection with most of the real jobs - 5065 classified correctly and a majority of the fake jobs - 834 caught. This balance of the false positive and negative supports trust and safety while minimizing unnecessary admin/human interventions.

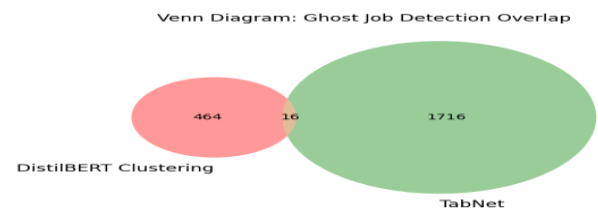
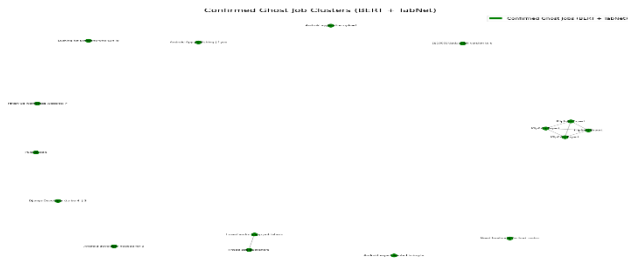


Fig. 7 Venn Diagram for Ghost Job Detection

The overlap analysis reveals in the venn diagram above **Fig. 7** shows 464 jobs flagged by clustering only, 1716 by TabNet only, and 16 by both which states that each method detects distinct fraud signals. This complementary approach enhances coverage by combining textual and structured insights.

6.2. The Business Value of the Findings



Ghost Job Clusters BERT+TabNet

The finding of the results helps us understand the importance of how textual and structured context-based learning can help us achieve a more sophisticated model, which provides deep insights into patterns of the fake jobs posting but also verify its credibility. By not using labeled dataset we are able to generalize the jobs posts enabling the model to understand and detect fake jobs and the coordinated network patterns it might be connected to. Unsupervised learning using TabNet Classifier helps the model use reconstruction error to find out anomalies further improving the generalization of the model.

These parameters allow to figure out clusters of fake job listings which is nearly identical but posted under different disguises, indicating coordinated fraudulent campaigns helping job seekers avoid such job offers in the future. By confirming ghost jobs through both textual and structured data anomalies, the model can proactively protect freelancers and help them not waste time or face scams which is especially valuable in regions where frauds are more susceptible. It also helps with operational efficiency by not relying solely on manual moderation, these graph-based insights can automate fraud detection pipelines, reduce human intervention load, and prioritize reviews based on cluster connections, saving significant time and cost. The cluster insights provided by the network classification will also help administrators create targeted rules for tracking fraud detection over time.

This clustering-based network solidifies the multi-angle detection strategy by adding interpretability and actionability. It converts predictions to real-world patterns that can be tackled with refined moderation, trust policies, or employer restrictions which offers efficient fraud protection that goes beyond black-box scores.

5. Conclusion

This project effectively shows a multi-angle approach for detecting fraudulent job listings on freelance platforms like Upwork. By combining semantic clustering using BERT embeddings with structured anomaly detection with the help of TabNet Classifier, we can uncover different scam patterns, including subtle text reuse and metadata inconsistencies for the jobs listed[6][7]. The low overlap between both methods (16 fake job records) highlights their complementary strengths, where both helps to identify

patterns from their respective area, reinforcing the value of a hybrid detection system.

Quantitative results such as high recall and a precision-recall trade-off, offers actionable insights for moderation without unfairly targeting legitimate users or vulnerable freelance job seekers. Visualization of confirmed ghost job clusters further validates real-world threats of coordinated fake job networks[1] and enables proactive fraud management. Overall, the solution not only helps in enhancing platform safety and user trust but also empowers administrators with a scalable, interpretable framework to tackle the ever evolving scam tactics in fake jobs scam.

6. References

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